Paper XII: Friction, Superconductivity, Plasma States, and Transport Mysteries: A Constraint-Driven Flux Map for Collective Matter

Steve Bico Mujjabi, MD
Independent Researcher
ORCID: 0009-0001-0556-5516

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Abstract

Paper XI extended CDFD into cosmological mysteries. Paper XII turns the same discipline toward laboratory-scale and engineering-scale mysteries: friction, stick-slip rupture, superconductivity, superfluidity, plasma confinement, magnetic reconnection, turbulence, jamming, glasses, and anomalous transport. These systems are not treated as solved by analogy. They are treated as falsifiable test beds for one claim: many physical mysteries arise where flux is forced through adaptive constraints whose capacity, memory, and topology change during transport. The paper separates established physics from CDFD interpretation, gives domain-specific mappings for Φ, C, and Ψ, states failure conditions for each proposed bridge, and includes reproducible outputs generated by the Paper XII supplementary script.

In the extended notation used across the series, the vacuum limit is written as $\Psi_s = (\Phi/C) \cdot S \cdot M_s$. In this manuscript the same notation is used cautiously as a transport map for measurable collective systems.

Contents

1 Purpose

The physics series now has two layers. Papers I–X contain the formal CDFD mass-spectrum and vacuum equation-of-state ladder. Paper XI is the speculative cosmology bridge. Paper XII adds a more testable bridge: condensed matter and collective media. Friction, superconductivity, plasma states, turbulence, and jamming are useful because they are measurable. They can be probed in the laboratory, simulated, and compared against mature standard theories of contact mechanics, superconductivity, vortex matter, transition to turbulence, and granular media [1, 2, 3, 4, 5]. This is also the scale where emergence is expected: collective systems can have organizing variables that are not visible in the microscopic description alone [6].

The manuscript therefore uses a strict posture:

1. conventional descriptions remain primary;
2. CDFD is introduced only as an organizing transport map;
3. every proposed bridge must name a measurable discriminant;
4. if the discriminant is not found after controlling for standard variables, the bridge fails.

2 Common Transport Form

Let a physical system be described by a driving flux Φ_X, a transport constraint C_X, and a local operating ratio

$$\Psi_{s,X} = \frac{\Phi_X}{C_X}. \tag{1}$$

For laboratory matter, C_X should not be read as a single universal substance. It is a domain-specific effective constraint: frictional contact strength, dephasing burden, disorder, magnetic confinement capacity, turbulent impedance, or jamming resistance.

The common adaptive law is

$$\partial_t C_X = \alpha_X |\Phi_X| + \eta_X M_X - \beta_X C_X + \gamma_X \nabla^2 C_X, \tag{2}$$

where M_X is a memory, damage, pinning, or ordering field. The added M_X term matters in this paper because frictional aging, vortex pinning, glass relaxation, and plasma hysteresis cannot be represented by a memoryless ratio.

2.1 Computational Status of This Paper

The figures in this manuscript are not empirical fits. They are deterministic toy outputs whose purpose is to make the proposed mappings explicit enough to fail. The generator `supplementary_XII.py` writes:

- a transport-mystery map linking each domain to its flux, constraint, transition signal, and failure condition;
- a stick-slip memory model with detected release events;
- a superconducting constraint-reallocation curve;

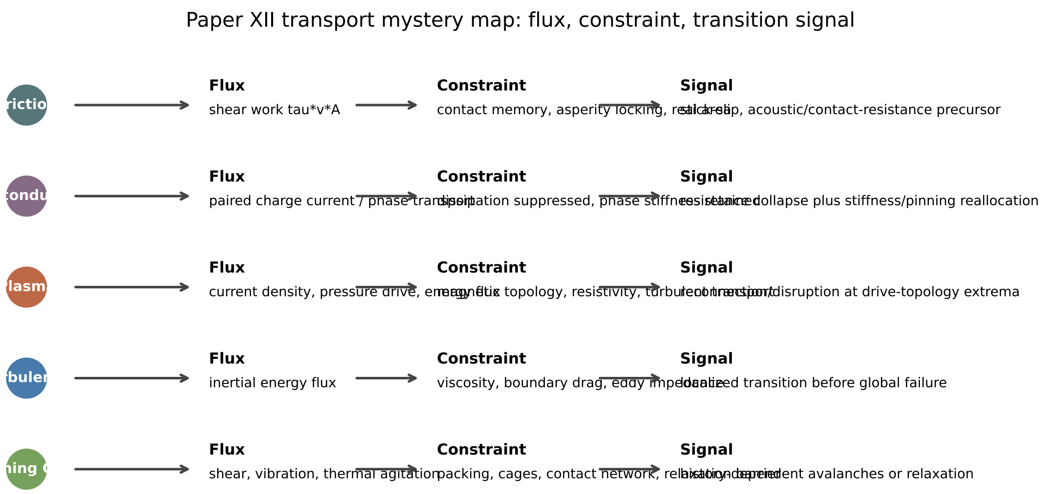

Figure 1: Generated Paper XII mystery map. The CDFD claim is not that one scalar replaces existing theories, but that many transitions are organized by adaptive transport constraints. The source table is `outputs/paper_XII/table1_transport_mystery_map.csv`.

- a plasma rupture-index field;
- a jamming memory distribution;
- symbolic and numerical gates checking that the examples behave as stated.

The outputs therefore serve as a scaffold for experiments, not as evidence that the bridge is already true.

3 Friction and Stick-Slip

Friction is often introduced as a force opposing motion, but real friction is not merely a number. It has memory, roughness, contact aging, heating, wear, acoustic emission, and abrupt slip. This makes it a natural CDFD target.

Let the shear flux be

$$\Phi_f = \tau v A, \tag{3}$$

where τ is shear stress, v is slip velocity, and A is the apparent contact area. Let the frictional constraint be

$$C_f = \mu_{\text{eff}} N A_r, \tag{4}$$

where N is normal load, A_r is real contact area, and μ_{eff} includes asperity chemistry, roughness, lubrication, temperature, and wear state.

Stick-slip occurs when the system crosses a constraint boundary:

$$\Psi_f = \frac{\tau v A}{\mu_{\text{static}} N A_r} \to 1^+. \tag{5}$$

During sticking, A_r and memory M_f can grow:

$$\partial_t M_f = a_f N - b_f v M_f - h_f T M_f. \tag{6}$$

During slip, the constraint partially collapses:

$$C_f^{\text{dynamic}} < C_f^{\text{static}}. \tag{7}$$

Hypothesis 1 (Friction as adaptive constraint memory)**.** *Frictional behavior is governed not only by instantaneous load and velocity, but by an adaptive constraint field whose memory variable controls the transition between static locking, dynamic sliding, and wear-mediated reorganization.*

Prediction 1 (Frictional precursor)**.** *Before macroscopic slip, acoustic emission, microscopic rearrangement, or contact-resistance noise should increase preferentially in regions where $\nabla(C_f^{-1}\nabla\Phi_f)$ is large, not merely where absolute load is maximal.*

Criterion 1 (Friction falsification)**.** *The CDFD friction bridge fails if stick-slip timing and localization are fully explained by standard rate-and-state or contact-mechanics variables with no residual dependence on a transport-memory field.*

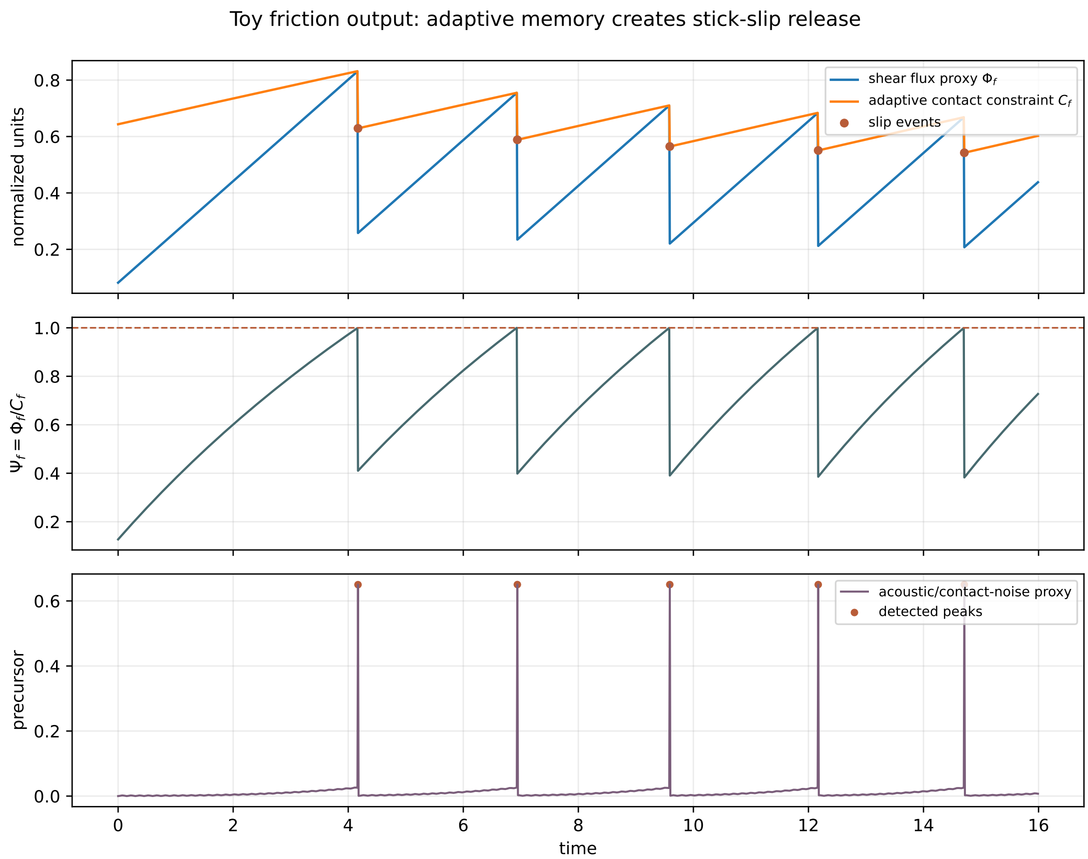

Figure 2: Generated stick-slip output. A slowly increasing shear-flux proxy loads an adaptive contact constraint until release. The lower panel shows a precursor/noise proxy. The event table is `outputs/paper_XII/table2_friction_stick_slip_events.csv`.

3.1 Why Friction Matters for CDFD

Friction is strategically important because it is the simplest everyday case where constraint is visibly real. A surface can be immobile under stress, then fail abruptly, then rebuild its contact structure. The same material pair may behave differently after waiting, heating, polishing, contaminating, or repeatedly sliding. That history dependence is exactly the kind of memory term represented by M_X in Eq. (2).

The CDFD addition to standard tribology is not a new coefficient of friction. It is a localization claim: slip should nucleate where flux, constraint weakness, and constraint gradients combine, not necessarily where load alone is largest. This gives the bridge a practical experimental target: synchronized contact imaging, acoustic emission, and local shear maps should identify whether $\nabla(C_f^{-1}\nabla\Phi_f)$ adds predictive value.

3.2 Triboelectricity

Triboelectric charging belongs in the same family. Contact and separation do not merely rub charges loose; they create and destroy interfacial transport channels. A CDFD formulation would track a charge flux Φ_q and an interfacial constraint C_q set by surface chemistry, humidity, roughness, trapped states, and separation rate. The useful claim is narrow: charge sign, magnitude, and relaxation should show path dependence on prior constraint history. If charging is completely predicted from instantaneous material pairs and environment, the CDFD memory bridge adds nothing.

4 Superconductivity and Superfluidity

Superconductivity is the cleanest place to avoid sloppy language. If C is called "constraint", one might incorrectly say superconductivity is $C \to 0$. That is not right. A superconductor has vanishing dissipative resistance, but it also has strong phase rigidity, quantized flux, coherence length, penetration depth, vortex pinning, and critical current.

Paper XII therefore separates two constraints:

$$C_{\mathrm{diss}} = C_{\mathrm{phonon}} + C_{\mathrm{disorder}} + C_{\mathrm{dephase}} + C_{\mathrm{vortex}}, \tag{8}$$

and

$$C_{\mathrm{phase}} = \rho_s, \tag{9}$$

where ρ_s is the superfluid or superconducting phase stiffness. The superconducting transport index is then

$$\mathcal{S}_{\mathrm{SC}} = \frac{\Phi_{\mathrm{pair}}\, C_{\mathrm{phase}}}{C_{\mathrm{diss}} + \epsilon}. \tag{10}$$

The superconducting transition is a regime change in which dissipative constraint collapses while phase constraint rises.

Hypothesis 2 (Coherence as constraint reallocation)**.** *Superconductivity is a constraint reallocation transition: dissipative scattering channels are suppressed, while phase stiffness and topological flux constraints become the dominant regulators of transport.*

This framing is compatible with conventional superconductivity only if it reproduces known limits: ordinary BCS superconductors, vortex dynamics in type-II materials, Josephson phase coherence, and critical current. It becomes interesting only where ordinary explanation is incomplete, especially high-T_c superconductors, strange metals, pseudogap regimes, and disorder-driven superconductor-insulator transitions.

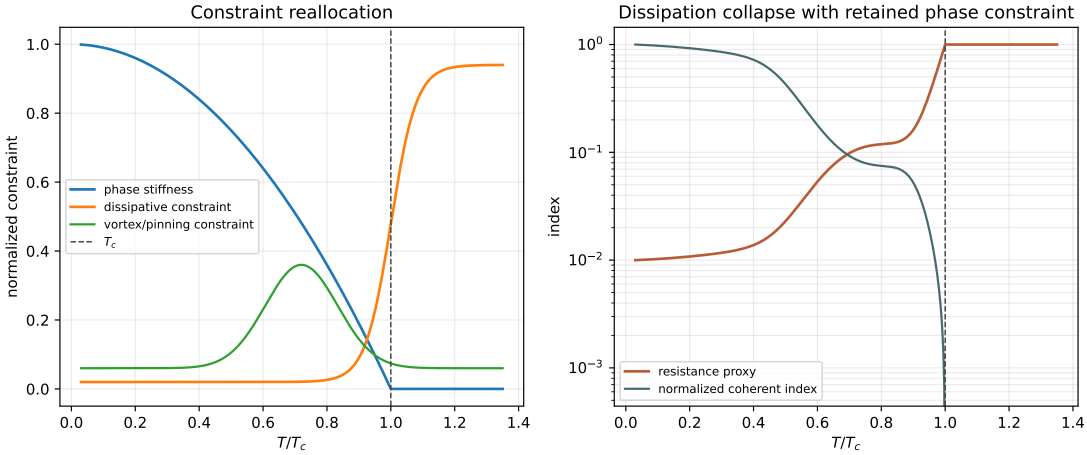

Figure 3: Generated superconductivity output. The toy model separates dissipative constraint from phase stiffness. Below the transition, dissipative resistance falls while phase/topological constraint remains active. The curve table is `outputs/paper_XII/table3_superconductivity_reallocation_curve.csv`.

4.1 Vortices and Pinning

In type-II superconductors, dissipation often appears through vortex motion. The relevant constraint is not ordinary resistance alone, but pinning topology:

$$C_{\text{vortex}} = C_{\text{pin}}(\mathbf{x}) - C_{\text{drive}}(J, B, T). \tag{11}$$

When current drive exceeds pinning capacity, vortices move and resistance returns. This gives a falsifiable CDFD statement: vortex avalanches and flux jumps should localize where the transport operator

$$\nabla \cdot (C_{\text{vortex}}^{-1} \nabla \Phi_J) \tag{12}$$

is extremal.

Prediction 2 (Superconducting discriminant)**.** *Across a superconductor-insulator transition, collapse of resistance should be accompanied by a measurable reallocation from dissipative impedance to phase stiffness or pinning topology. If resistance falls without any measurable change in coherence, stiffness, or topological constraint, the CDFD interpretation is weakened.*

Criterion 2 (Superconductivity falsification)**.** *The superconductivity bridge fails if it cannot recover standard superconducting constraints and if its added variables do not improve prediction of high-T_c, vortex, or disorder-driven anomalies.*

4.2 Superfluidity

Superfluidity is the neutral analogue: mass flow proceeds without ordinary viscous dissipation, but quantized vortices and phase coherence still constrain transport. The CDFD reading is again not "no constraint". It is a switch from dissipative constraint to topological phase constraint.

4.3 High-T_c and Strange-Metal Boundary

High-T_c superconductivity is where Paper XII could eventually matter most. Conventional superconductivity already has a mature theory. The difficult territory is where pairing, phase stiffness, disorder, stripe order, spin fluctuations, pseudogap behavior, and strange-metal transport do not collapse into one simple transition.

In CDFD language, this is a decoupling problem:

$$C_{\text{pair}} \neq C_{\text{phase}} \neq C_{\text{diss}}. \tag{13}$$

Pair formation may begin while phase coherence remains constrained; dissipation may remain anomalous even when a partial gap exists. A useful CDFD model must therefore predict which constraint collapses first and which remains active. If it cannot distinguish pairing onset from phase locking, it adds no value.

5 Plasma States and Magnetic Reconnection

Plasma systems already fit CDFD language naturally. A plasma is a medium where charged-particle flux, magnetic topology, pressure gradients, turbulence, and radiation losses compete. In a confinement device, solar corona, or magnetosphere, the relevant variables are:

$$\Phi_p \sim \text{current density, pressure-gradient drive, or energy flux,} \tag{14}$$

$$C_p \sim \text{magnetic topology, confinement capacity, resistivity, and turbulent transport.} \tag{15}$$

The public model used here states only a scientific rupture criterion: plasma instability is expected when a high-Ψ drive coincides with steep constraint gradients. In this manuscript the criterion is written as

$$\mathcal{R}_p = |\nabla C_p| \frac{\Phi_p}{C_p} \tag{16}$$

is a candidate rupture index for reconnection, edge-localized modes, or disruption-like events.

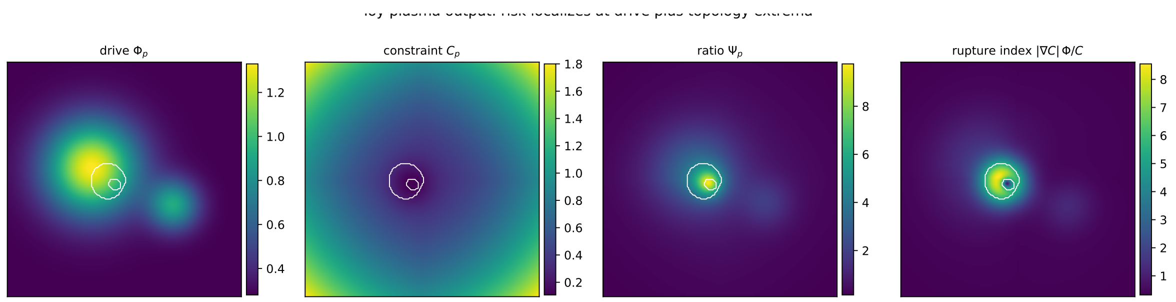

Figure 4: Generated plasma output. The rupture index combines drive, constraint, and constraint-gradient information. In the toy field, risk does not follow the drive field alone. The summary table is `outputs/paper_XII/table5_plasma_rupture_index_summary.csv`.

Hypothesis 3 (Plasma rupture as constraint-topology failure)**.** *Plasma disruptions and reconnection events occur where current or pressure-gradient flux exceeds the local capacity of magnetic topology to preserve confinement.*

Prediction 3 (Plasma precursor)**.** *Before a disruption, the highest-risk region should be better predicted by a combined drive-plus-topology index such as $\mathcal{R}_p$ than by absolute plasma energy density alone.*

Criterion 3 (Plasma falsification)**.** *The plasma bridge fails if disruptions, reconnection, or confinement loss correlate only with standard scalar plasma parameters and not with constrained transport geometry after controlling for known MHD variables.*

5.1 Plasma States as CDFD Regimes

Regime	**CDFD reading**	**Observable handle**
Diffuse plasma	low coherence, weak constraint topology	broad particle/field distribution
Confined plasma	$\Psi_p \approx 1$ with magnetic topology sustaining flux	long confinement time, low loss rate
Reconnection	steep constraint gradient breaks and reroutes flux	rapid magnetic-energy release
Runaway/disruption	sustained $\Psi_p > 1$ over critical region	quench, edge loss, global instability
Ignition-like state	self-heating flux maintains organized constraint	sustained fusion-relevant balance

6 Turbulence, Drag, and Boundary Layers

Turbulence is the classical transport mystery. A laminar flow has organized flux and stable constraint. A turbulent flow reorganizes energy across scales. CDFD should not pretend to replace Navier–Stokes. Its role is to provide a phase-structure language:

$$\Psi_{\text{fluid}} = \frac{\Phi_{\text{inertial}}}{C_{\text{viscous}} + C_{\text{boundary}} + C_{\text{eddy}}}. \tag{17}$$

The denominator is intentionally adaptive: once turbulence begins, eddy structure becomes a new transport constraint rather than a passive consequence.

Hypothesis 4 (Turbulence as adaptive constraint production)**.** *Turbulence is a regime in which inertial flux produces its own multiscale constraint field, redistributing transport through eddies, boundary layers, and coherent structures.*

Prediction 4 (Transition localization)**.** *The laminar-to-turbulent transition should localize where boundary-layer constraint cannot dissipate inertial flux smoothly, not simply where global Reynolds number is high.*

Criterion 4 (Turbulence falsification)**.** *The turbulence bridge fails if no local constraint-memory or geometry variable improves transition prediction beyond standard hydrodynamic variables.*

6.1 Drag Reduction and Flow Control

Drag reduction is a practical reason to keep turbulence in the mystery queue. Polymers, riblets, compliant walls, bubbles, and active flow-control systems all alter the way flux encounters boundary constraint. CDFD predicts that successful drag reduction should not merely lower average resistance; it should change where constraint gradients form and how coherent structures reroute momentum.

This is experimentally concrete. Particle-image velocimetry and wall-shear sensors can compare three fields: inertial flux, boundary constraint, and coherent-structure transport. If drag reduction works while those fields remain unchanged except for a scalar viscosity shift, the CDFD interpretation is weak. If it works by redistributing constraint topology, the bridge becomes more interesting.

7 Glasses, Jamming, Granular Flow, and Avalanches

Glasses and jammed materials are almost the inverse of superconductors. Superconductors suppress dissipative constraint and permit coherent transport. Jammed systems accumulate constraint faster than flux can relax it.

For a granular or glassy material:

$$\Phi_g \sim \text{imposed shear, vibration, or thermal agitation,} \tag{18}$$

and

$$C_g \sim \text{cage topology, packing fraction, contact network, and relaxation barrier.} \tag{19}$$

The jamming boundary is

$$\Psi_g = \frac{\Phi_g}{C_g} \ll 1 \tag{20}$$

for trapped, over-constrained states, while avalanche failure occurs when stored stress abruptly exceeds local constraint:

$$\Psi_g^{\text{stored}} \to 1^+. \tag{21}$$

Prediction 5 (Jamming memory)**.** *Two samples with the same density and shear rate but different preparation histories should show different avalanche statistics if the CDFD memory field M_g is real.*

Criterion 5 (Jamming falsification)**.** *The jamming bridge fails if preparation history, contact-network memory, and constraint topology add no predictive information beyond instantaneous density, stress, and temperature.*

8 Fracture, Cracks, and Material Failure

Fracture is friction's irreversible cousin. A crack is a transport event: elastic energy flux localizes into a narrowing process zone while material constraint weakens at the crack tip. In a CDFD representation:

$$\Phi_{\text{fracture}} \sim \text{elastic energy release rate,} \tag{22}$$

$$C_{\text{fracture}} \sim \text{local toughness, plastic dissipation, microstructural barriers.} \tag{23}$$

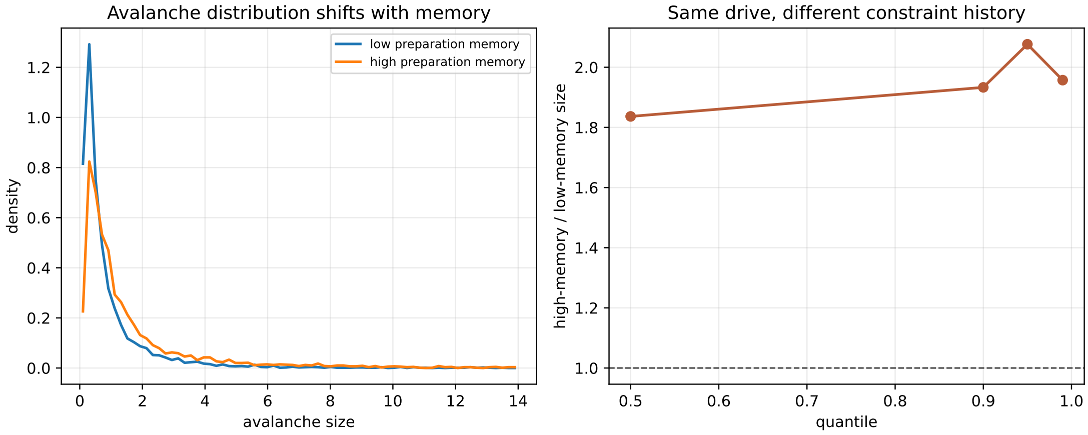

Figure 5: Generated jamming output. Two systems with the same nominal drive but different preparation memory produce different avalanche tails. The distribution table is `outputs/paper_XII/table6_jamming_memory_distributions.csv`.

The crack path should follow corridors where stored energy flux overwhelms local material constraint:

$$\mathcal{K}_{\mathrm{CDFD}} = \nabla \cdot \left(C_{\mathrm{fracture}}^{-1} \nabla \Phi_{\mathrm{fracture}} \right) . \tag{24}$$

This does not replace stress-intensity factors or fracture mechanics. It asks whether heterogeneous fracture paths, branching, and avalanche acoustic emission are better predicted when constraint topology is explicitly mapped.

Prediction 6 (Fracture corridor)**.** *In heterogeneous materials, crack branching should align with high $\mathcal{K}_{\mathrm{CDFD}}$ corridors after controlling for ordinary stress intensity.*

Criterion 6 (Fracture falsification)**.** *The fracture bridge fails if crack paths and acoustic-emission bursts are fully predicted from standard fracture variables with no residual role for adaptive constraint topology.*

9 Topological and Field-Written Matter

Several modern materials are controlled by topology or externally written constraint fields. Quantum Hall systems, topological insulators, ferrofluids, magnetorheological fluids, and vortex matter in superconductors all show transport that is not just local resistance. Boundaries, defects, and applied fields determine which channels are available.

The CDFD bridge here is:

$$C_{\mathrm{top}}(\mathbf{x}, t) = C_{\mathrm{bulk}} + C_{\mathrm{boundary}} + C_{\mathrm{defect}} + C_{\mathrm{field}}. \tag{25}$$

If topology protects a channel, the effective constraint along that channel is low while the surrounding bulk constraint remains high. If an external magnetic field writes chains in a ferrofluid, it changes C_{field} and therefore the macroscopic flow response.

The key falsification test is again incremental prediction. If a topological invariant, field geometry, or defect map already predicts the transport completely, CDFD is only a translation. If the adaptive constraint law predicts transition timing, hysteresis, or failure of protection, it earns a place.

10 Additional Mystery Queue

The following topics are worth keeping in the expandable spine. Some should remain short subsections until there is enough mathematics or data for a standalone paper.

Mystery	CDFD bridge	Needed discriminant
High-T_c superconductivity	phase stiffness and dissipative constraint decouple before full coherence	distinguish pairing scale from phase-locking scale
Strange metals	charge flux lacks ordinary quasiparticle constraint	transport scaling tied to adaptive scattering field
Quantum Hall and topological phases	transport protected by topological constraint channels	edge-state resilience predicted by constraint topology
Sonoluminescence	collapse focuses hydrodynamic flux through transient constraint shell	emission tied to maximum compression topology
Lightning and ball-lightning candidates	atmospheric charge flux constrained by plasma channel memory	persistence depends on channel constraint lifetime
Non-Newtonian fluids	imposed shear modifies the constraint field itself	shear-thickening/thinning predicted by $C(\dot{\gamma})$ dynamics
Fracture and crack propagation	stress flux localizes along weakening constraint corridors	crack path follows high-$\nabla(C^{-1}\nabla\Phi)$ regions
Magnetorheological and ferrofluids	external field writes transport constraint topology	viscosity/yield stress follows field-imposed C geometry

10.1 Priority Ranking

Not all mysteries should be expanded equally. The best next candidates are the ones where data can be produced or found quickly:

1. **Friction and fracture**: accessible laboratory systems with imaging and acoustic signals.
2. **Plasma**: existing confinement and reconnection diagnostics can test drive-plus-topology predictors.
3. **Superconductivity**: harder, but strategically powerful because CDFD must distinguish phase stiffness from dissipation.
4. **Jamming and glasses**: excellent for memory and preparation-history tests.
5. **Lightning and sonoluminescence**: interesting but should stay exploratory until the variables are better pinned down.

11 Vacuum-Memory and Extreme-Field Experimental Roadmap

Paper I introduced M_s as a possible memory or recovery state of the vacuum medium. That idea is not established by the numerical recovery of $\chi = 1/\alpha$. It becomes scientific only if it creates discriminants that can be compared against QED, general relativity, and established laboratory-domain models. The tests below are therefore written as falsification targets, not as claimed observations.

11.1 Target parameters from the source notes

The DOCX source notes singled out these experiments as high-priority tests. This manuscript keeps descriptive labels, because the scientific content is the measurable discriminant. The common parameter is the memory relaxation rate

$$\lambda_M = \tau_M^{-1}, \tag{26}$$

where τ_M is the vacuum-memory recovery time proposed in Paper I. No numerical value is assigned here. A value or bound must come from either a deeper vacuum equation of state or a controlled null/result in the experiments below.

Test family	**Primary target**	**Required control**
Pump–probe vacuum hysteresis	history-dependent phase residual decaying with τ_M	nonlinear QED, plasma, gas, optics, thermal, and detector effects
Extreme-field spectroscopy	residual alpha-sensitive transition shift scaling with flux intensity	standard QED, nuclear structure, thermal, magnetic, and systematic corrections
Rotating-source residual	spin-correlated pressure-gradient signal	Newtonian gravity, vibration, magnetic coupling, electrostatics, air drag, and motor heating

11.2 High-frequency vacuum hysteresis

Let a high-field pulse produce a temporary overload functional

$$\mathcal{H}(t) = \int_\Omega \max\left(0, \frac{|\boldsymbol{J}(\boldsymbol{x}, t)|}{C(\boldsymbol{x}, t)} - 1\right) dV. \tag{27}$$

If the memory kernel proposed in Paper I is physical, a second probe pulse sent before the relaxation time τ_M has elapsed should see a residual response that depends on the history of the first pulse:

$$\Delta\phi_2 = F(I_1, \Delta t, \tau_M) - F(0, \Delta t, \tau_M), \tag{28}$$

where I_1 is the pump intensity and Δt is the pump–probe delay.

Prediction 7 (Vacuum hysteresis). *After all known nonlinear QED, plasma, gas, mirror, thermal, and detector effects are controlled, a nonzero history-dependent phase residual should scale with pump intensity and decay with a finite relaxation time if M_s is a real vacuum state variable.*

Criterion 7 (Vacuum-hysteresis falsification). *The vacuum-memory bridge fails if high-field pump–probe experiments find no history-dependent residual beyond known QED and instrumental effects, or if any residual does not scale with the overload functional $\mathcal{H}$.*

11.3 Extreme-field spectral drift

The strongest publishable form of this idea is narrow. If $\alpha = a/R$ is an adaptive fixed point rather than a rigid input, extreme field environments could in principle create a tiny effective displacement:

$$\frac{\Delta\alpha}{\alpha} = \mathcal{F}\left(\frac{\Phi}{C}, S, M_s\right), \qquad \mathcal{F}(0,1,1) = 0. \tag{29}$$

The natural search space is high-precision spectroscopy of highly charged ions, strong-field QED systems, astrophysical spectra, or other environments where standard calculations already predict the dominant shifts.

Prediction 8 (Extreme-field alpha sensitivity). *If adaptive vacuum response is real, transition energies that depend strongly on α should show a small residual correlated with controlled field intensity or environmental flux after all standard QED, nuclear, thermal, and systematic corrections are applied.*

Criterion 8 (Spectral falsification). *The bridge fails if high-precision spectra continue to match standard QED and known systematics with no residual scaling with Φ/C, or if any residual is not reproducible across independent platforms.*

11.4 Rotating-mass pressure-gradient test

The gravitational extension of CDFD is the weakest of the three vacuum-level targets and should be treated accordingly. The hypothesis is that rapidly organized angular momentum in dense matter may produce a secondary pressure or constraint-gradient contribution:

$$\Delta g_{\mathrm{CDFD}} \propto \nabla\left(C_{\mathrm{vac}}^{-1}\Phi_{\mathrm{rot}}\right). \tag{30}$$

A defensible experiment would require shielded, balanced rotors, blind reversal of spin direction, vibration isolation, magnetic/electrostatic controls, and an atom-interferometric or torsion-balance readout.

Prediction 9 (Rotating-source residual). *After conventional Newtonian, relativistic, vibration, thermal, electromagnetic, and air-coupling effects are subtracted, any remaining spin-correlated signal should scale with angular momentum and source geometry, not merely with motor power or mechanical noise.*

Criterion 9 (Rotating-source falsification). *The rotating-mass bridge fails if controlled experiments find no spin-correlated gravitational residual, or if observed residuals track known engineering systematics rather than the predicted constraint-gradient geometry.*

11.5 Experimental priority

The priority order should be conservative:

1. high-frequency vacuum hysteresis, because it directly targets M_s;
2. extreme-field spectroscopy, because it tests whether α behaves as a rigid constant or an adaptive fixed point;
3. rotating-source gravity, because the expected signal is most vulnerable to mundane systematics and therefore needs the strictest controls.

None of these tests is evidence by itself. Their value is that they make the non-equilibrium extension of Paper I falsifiable.

12 Unified Falsification Program

Domain	**Primary measurement**	**Failure condition**
Friction	contact-resistance noise, acoustic emission, real contact area	no residual memory/geometric predictor after rate-and-state modeling
Superconductivity	phase stiffness, resistance, vortex motion, disorder	CDFD variables do not improve transition or vortex predictions
Plasma	current profile, magnetic topology, turbulence, confinement time	disruptions depend only on known scalar parameters
Turbulence	local Reynolds fields, boundary-layer stress, coherent structures	no local constraint field improves transition localization
Jamming/glass	packing, contact network, relaxation time, avalanche sizes	preparation history adds no predictive information

The correct standard is incremental predictive value. CDFD does not win by renaming known physics. It wins only if the Φ/C mapping and adaptive-constraint law predict where a transition occurs, how it scales, or why two apparently similar systems diverge.

13 Reproducibility Artifacts

Paper XII now has its own output set:

Artifact	Purpose
fig12_0_transport_mystery_map.pdf	visual flux/constraint map for the included mysteries
fig12_1_friction_stick_slip.pdf	adaptive-memory stick-slip toy model
fig12_2_superconductivity_reallocation.pdf	dissipative vs phase-constraint transition model
fig12_3_plasma_rupture_index.pdf	drive/topology rupture-index field
fig12_4_jamming_memory.pdf	preparation-memory avalanche distribution
checks_summary_XII_gate.csv	pass/fail summary for generated outputs

The generator is intentionally modest. The outputs are not evidence for the physical theory; they are a precise statement of what the proposed theory means operationally. This matters because a falsifiable wrong model is more useful than an unfalsifiable analogy.

14 Relation to Papers I–XI

Paper I introduced the physical-vacuum intuition using vortex stability. Papers II–X developed the mass-spectrum and vacuum-EOS ladder. Paper XI used that ladder to frame cosmological hypotheses. Paper XII is different: it is a proposed laboratory bridge. It asks whether the same flux-constraint vocabulary can organize measurable collective states without leaning on cosmology.

The strongest reason to keep Paper XII is strategic. A reader may reject the dark-sector bridge, but still test frictional memory, vortex pinning, plasma rupture, or jamming as constrained transport. These are nearer to experiment, easier to falsify, and useful for sharpening the whole theory.

15 Conclusion

Friction, superconductivity, plasma confinement, turbulence, and jamming are not random additions to the physics series. They are the everyday places where transport mystery is most visible. A surface locks then slips. A material loses resistance but gains phase rigidity. A plasma confines then reconnects. A fluid flows smoothly then becomes turbulent. A dense material behaves solid until it avalanches. In CDFD language, these are all candidates for adaptive constraint transitions.

The claim remains conditional. If the measurable flux and constraint fields do not improve prediction beyond established models, Paper XII is only a metaphor. If they do, these laboratory mysteries become the bridge between the formal CDFD vacuum papers and practical physics.

Acknowledgments

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